

Recognition of Heart Disease based on Primitive Parameters using Machine Learning Model

Jyothirmai Digumarthi,
Department of Networking and
Communications , Faculty of
Engineering and Technology, SRM
Institute of Science and
Technology, Kattankulathur,
Tamilnadu, India..
Email: jd6957@srmist.edu.in

*V. M. Gayathri,
Department of Networking and
Communications, Faculty of
Engineering and Technology, SRM
Institute of Science and
Technology, Kattankulathur,
Tamilnadu, India.Email:
gayathrm@srmist.edu.in

R. Pitchai,
Department of Computer Science
and Engineering,
B V Raju Institute of Technology
Narsapur,Telangana State, India
Email: pitchrks1984@gmail.com

Abstract - Cardiovascular diseases (CVDs) have a significant worldwide impact, affecting millions of victims. The complex nature of these conditions requires modern treatments to address them effectively. In order to forecast certain diseases, machine learning algorithms take into account factors like age, gender, and blood pressure, amongst other characteristics. To enhance the usefulness of these algorithms, it is necessary to carefully choose their characteristics in an ideal manner. Various techniques, in order to identify the necessary traits, many techniques, including Recursive Feature Elimination, the least absolute shrinkage and selection operator, Relief approach, and mutual data, are being applied. Methods from the field of machine learning (ML) such as random forest, logistical regression, and K-nearest neighbours (KNN) are frequently employed in the process of predicting cardiovascular disease (CVD). These approaches have demonstrated beneficial effects and may be used to enhance the health of patients through early diagnosis. This study proposes an integrated approach that blends feature selection approaches with machine learning techniques in order to enhance cardiovascular disease prediction. Several methodologies have been evaluated in order to determine the most effective approach for predicting cardiovascular illnesses.

Keywords—Cardiovascular Diseases, Machine Learning, Feature Selection, Random Forest, K-Nearest Neighbors, Logistic Regression (LR).

I. INTRODUCTION

In 2008, a survey revealed that around 17.3 million individuals' lives were claimed by cardiac diseases corresponding to 30% of all global deaths. Accurate prediction of cardiovascular diseases (CVDs) is essential to ensure patient safety. Predictive machine learning techniques, when given the appropriate information, demonstrate significant results in identifying and predicting CVDs. Table 1 details the common risk factors of heart diseases.

TABLE 1: COMMON RISK FACTORS OF HEART DISEASES

Factor	Explanation
Poor Diet	Eating the right foods is necessary for a healthy life and helps avoiding threatening diseases like CVDs.
Physical Inactivity	Absence of physical movement can lead to high blood pressures causing heart diseases. It can also cause the development of coronary artery disease

Smoking	Chemicals exposed to via smoking can damage blood cells and affects the proper functioning of heart eventually leading to heart attacks
Obesity	Excessive weight disrupts the proper functioning of the left ventricle due to insufficient filling of blood between heart beats.
Hypertension	Hypertension or high blood pressure increases risk of walls getting stretched and damaged over time.
High Blood Cholesterol	Presence of cholesterol narrows the walls of the arteries, restricting blood flow.

Methods from the field of machine learning (ML), including Logistic Regression, Random Forests, and k-nearest-neighbors (KNN), are utilized in the process of accurately diagnosing cardiac conditions. While KNN relies on the nearest neighbor, RF relies on the decision tree notion. When calculating binary probabilities, LR is used. The algorithm's performance is impacted by the quality of the data. In order to increase ML effectiveness, it is critical to choose the appropriate characteristics. Techniques like LASSO and relief optimization can be applied to CVD forecasting.

II. LITERATURE REVIEW

The researchers in this study conducted a comparative analysis of many algorithms, evaluating their performance based on accuracy and various other parameters. The researchers discovered that the Support Vector Machine technique had superior performance in terms of accuracy and efficiency compared to other competing methods. The outcomes of the research indicate that the proposed AI model has the potential to serve as a dependable tool for diagnosing heart diseases.

Cheng et al. (2012) proposed the application of a machine learning algorithm-based artificial intelligence technique for the detection of heart disease. The dataset comprises 303 patients' information, each characterized by 14 characteristics. Multiple approaches were used to construct the model. The researchers assess the efficacy of various approaches via the use of diverse criteria. The results indicate that, in relation to performance measures, the Support Vector Machine Algorithm outperforms alternative techniques.

In their study, Abdallah Abdellatif et al. (2019) conducted a comparative analysis of two deep learning models in order to assess their efficacy in the diagnosis of heart disease. One of the models used a sequential feature selection technique, whilst the other model refrained from using any kind of feature selection. The research revealed that the model using successive feature selection had superior performance compared to the model that did not use feature selection, as shown by higher levels of accuracy and precision. The research findings indicate that the use of sequential feature selection techniques may effectively improve the ability of Machine Learning Models to accurately identify and diagnose Cardiac illness.

Ahmad et al. (4) have presented an artificial intelligence (AI)-based approach for identifying and categorizing the severity of cardiovascular diseases (CVDs). The model developed with machine learning methods and a dataset of 303 patients, each characterized by 13 distinct features. The authors also used hyperparameter optimization techniques in order to enhance computing performance. According to the findings of the research, the framework has the ability to become a viable instrument for the early diagnosis of cardiac disease as well as the classification of the disease's severity level.

Within the scope of e-healthcare, the research conducted by Hira Fatima and colleagues [5] utilised an approach predicated on machine learning for the purpose of identifying cardiovascular illnesses. For the purposes of both training and testing the model, the research utilised a dataset consisting of 303 patients and 13 different variables. The efficiency of a large number of algorithms was studied, and the method that proved to be the most effective in terms of both its precision and other pertinent performance metrics was selected. According to the findings of the investigation, the model that was developed has a respectable degree of accuracy when it comes to identifying cardiac incidents among people. The results of the study showed that the proposed approach is likely to be a reliable strategy for the timely diagnosis of cardiovascular sickness in e-healthcare systems. This was proved by the findings of the study.

In their study, Santosh et al. (2016) did a comprehensive literature analysis focusing on machine learning techniques used in the field of gender and age determination using X-ray images of teeth. The authors conducted an in-depth review of notable scholarly works and delineated the advancements in this field in their publication, which was disseminated in the Journal of Healthcare Engineering in the year 2022. The insights provided by the survey on different algorithms and methodologies might be of considerable help to researchers engaged in the study of dental image analysis. Sujata et al. (7) The current study investigates a number of different machine learning approaches that can be utilized for the forecasting of cardiovascular illnesses. The models were created with the use of several different methodological approaches, and the dataset included 303 individuals with 14 characteristics. The authors evaluate the effectiveness of a variety of algorithms by employing accuracy and numerous other pertinent parameters in their analysis. According to the findings, the approach known as Random Forests beats other approaches in terms of metrics such as efficiency and accuracy. The outcomes of the research provide support to the proposition that machine

learning algorithms may serve as a dependable tool for detecting cardiac events.

The research done by Ghosh et al. (2018) demonstrates that the implementation of attribute selection techniques has a substantial impact on the efficacy of models. This suggests that the suggested technique might be regarded as a dependable method for detecting early onset of cardiovascular disease (CVD). The present study used a dataset including 303 patients and encompasses 14 unique variables. The evaluation of the methodology was conducted by using several machine learning techniques.

The research that was carried out by Jian Ping and colleagues [9] looks on the feasibility of employing lasso and relief techniques that are used in machine learning algorithms to make predictions regarding the risk of coronary artery disease. In order to determine how successful their strategy is, the researchers use three different Machine Learning approaches to a dataset that contains 303 individuals and 14 traits. According to the findings of the research, implementing LASSO decision-making procedures into model development results in significant enhancements in the models' overall performance. As a result, the findings of the study indicate that this method has the potential to be regarded as a trustworthy instrument for early detection of cardiovascular disease.

In the research they conducted, Senthilkumar Mohan and colleagues (10) employed four different machine-learning approaches to conduct an analysis on a dataset that included 303 patients and 14 parameters. Based on the performance tests conducted, the authors demonstrate that their hybrid strategy, which integrates two algorithms, has superior performance compared to the separate approaches. The findings provide support to the notion of using hybrid technology in the identification of heart ailments.

In 2020, Ibrahim SS et al. [11] announced a cardiovascular disease (CVD) prediction technique that utilizes an evolutionary memetic weighted association categorization approach. The present publication provides a comprehensive overview of the algorithm used in the research and endeavors to explore relevant literature, methodologies, and potential future approaches within the domain of illness prediction.

Akbas et al. (2012) conducted a study. In order to assess the pertinence and reliability of the identified rules in the cardiac dataset, The authors centred their attention on making use of association rules and incorporating the assurance component. The purpose of the research was to improve the efficacy of rule-based models for the diagnosis and prognosis of cardiovascular disease.

The objective of the study carried out by Bashir et al. (2013) was to improve the precision of heart attack forecasting systems by employing a variety of feature selection strategies in order to extract relevant data from the dataset. The purpose of this study was to investigate the effect that selecting certain features has on the effectiveness and understandability of machine learning models with predictive capabilities that are employed in the diagnosis and prognosis of heart disease.

Kharyaet et al. (2014) developed a weighted Bayesian rule mining association strategy with the intention of producing Bayesian Belief Networks utilizing an association rules-

based methodology that they proposed. By including weights into the rule mining process, the purpose of the research was to improve the accuracy of Bayesian Beliefs Networks as well as the interpretability of the results of the analysis. This was done in order to improve the capabilities of the final network with regards to representation and inference.

III. ENHANCED PREDICTION APPROACH

In recent years, there has been an uptick in interest surrounding the advancement of predicting heart disease through the application of algorithms for machine learning and feature selection strategies. This interest has come about as a result of a rise in the prevalence of heart disease. The use of this technology has resulted to an enormous boost in the accuracy of the diagnosis of cardiovascular disease, with present rates varying between 83.5 percent to 98.1 percent accuracy. These enhancements have been achieved via the use of machine learning methods. This recent effort offers a notable divergence from previous models, which exhibited decreased precision as a result of inadequate feature selection and inefficient Machine Learning methodologies. The purpose of this inquiry is to evaluate the predicted efficacy of three artificial intelligence algorithms, specifically Logistic Regression, K-Nearest Neighbors, and Random Forests, in the context of heart disease prediction. Furthermore, one of our primary goals is to evaluate the extent to which the precision of these machine-learning models may be enhanced by the use of diverse feature selection techniques. The primary objective is to create an advanced machine-learning model that can accurately predict heart diseases using a restricted set of input variables.

IV. METHODOLOGY

The methodology in this investigation aims to comprehend the effectiveness of the ML algorithms: RNN, RF and logistic regression in predicting cardiac events. The below actions were undertaken to fulfill the research goals:

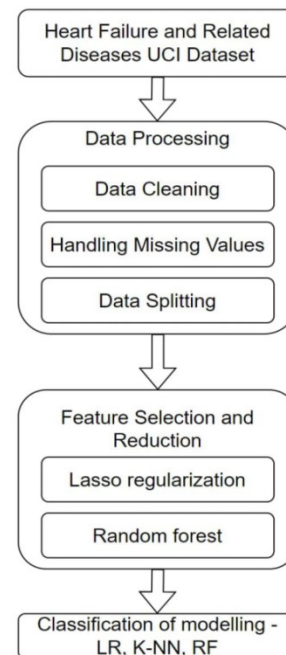


Fig.1 Methodology

A. Proposed Work

Machine learning algorithms have the power to predict the likelihood of cardiac events in people, hence enabling the provision of suitable preventive care measures to mitigate the risk of death. The objective of this study is to use machine learning methodologies, namely random forest, logistic regression, and K-nearest neighbors (KNN), to make predictions about cardiovascular diseases (CVDs). Figure 1 provides a comprehensive depiction of the proposed process.

B. Dataset Description

Heart diseases and related conditions the 76 properties of the UCI dataset are utilized, with 14 attributes - such as Blood Pressure, age, Sex, and symptoms- getting the most attention. Data is split into 30 percent for testing and 70 percent for training. With few input characteristics, the goal is to create a model with high accuracy using techniques like forward choosing, correlation-based analysis, and backward removal to extract the most crucial features.

C. Dataset Preprocessing

The prediction process relies significantly on the collection of data for machine learning algorithms such as Random Forest (RF), K-Nearest Neighbors (KNN), and Logistic Regression (LR). Commonly used preprocessing methods include data cleaning, feature scaling, handling missing values, and dataset split.

- **Data Cleansing:** To ensure data reliability, elimination of duplicates must be done. Outliers, unusually high values that can negatively impact performance should be dealt with help of statistical procedures. Make necessary updates after thoroughly checking the data for consistency and accuracy.
- **Handling Unavailable Values:** Identifying missing values in dataset and resolving them with caution is vital as they can affect the algorithm's performance. This can be done through various methods such as mean imputation, median

imputation, mode imputation, KNN imputation and regression imputation.

- Data splitting: From the preprocessed data, generate the testing and training datasets. The testing dataset is utilized to evaluate the model trained using the training dataset. Usually, 20-30% is used for testing and 70-80% is used for training depending on requirements and dataset size.

D. Feature Selection and Reduction:

Feature selection is the process of choosing and excluding certain features without making any changes to them. Dimensionality reduction is a technique that involves transforming the characteristics of a dataset into a lower-dimensional space. Table 2 Categorize Heart Disease Prediction System's important characteristics.

TABLE 2: CLINICAL CHARACTERISTICS FROM THE UCI DATASET UTILIZED TO CATEGORIZE HEART DISEASE PREDICTION SYSTEMS

Characteristics	Illustration
Trestbps	Resting Blood Pressure
Chol	mg/dl of Serum cholesterol.
Exang	A boolean measurement showing whether exercise-induced angina has been experienced by the patient.
Fbs	Whether the fasting blood sugar is more than 120 md/dl is shown by a boolean value (0=false; 1=true).
Age	In years old.
Thal	The condition of the heart, which may be described as normal, fixed defect, or reversible defect.
Sex	Female, male
Predicted attribute	Either 0 or 1
Ca	The number of main vessels that are coloured by fluoroscopy, ranging from 0 to 3.
Cp	Chest pain type 1: non-anginal; 2: classic angina; 3: atypical; 4: asymptotic
Restecg	Results of Electrocardiogram during rest
Oldpeak	ST depression as a result of activity in comparison to rest
Thalach	Maximum heart rate attained
Slope	The rate of change of the ST segment during maximal physical activity.

E. Classification Modeling

- Logistic regression is a versatile statistical technique used in academic research and professional practices. The technique is capable of processing both category and numerical data, accommodating binary as well as multi-class settings. The process of data preparation includes the handling of variable encapsulation problems

and the scaling of characteristics, hence improving the overall dependability of the data.

- The K-Nearest Neighbours algorithm is a popular machine-learning technique used for classification and regression tasks. It is a non-parametric method that makes predictions based on the K-Nearest Neighbours (k-NN) algorithm is a non-parametric classification strategy that allocates a class to a new instance by considering the consensus of its k-nearest neighbors. The proposed methodology is capable of effectively addressing both multi-class and binary classification issues. It does this by using both category and numerical features in its analysis. Additionally, the strategy uses the Euclidean distance metric to measure the similarity of instances. Data preparation is an essential step in the data analysis process, including various ways to handle missing data, encode variables, and scale features. By modifying the value of parameter k, the algorithm may effectively capture local patterns while also taking into account broader trends, hence leading to enhanced performance. The K-Nearest Neighbors (K-NN) algorithm is widely recognized as a helpful tool in research owing to its robust classification performance.
- The Random Forest method is well acknowledged in the field of machine learning for its remarkable prediction powers. This specific methodology is often used in scholarly investigations. The technique involves the integration of numerous decision trees to get reliable results. The proposed approach demonstrates proficiency in handling both category and numerical features, successfully tackling issues related to multi-class and binary classification. Optimal outcomes are produced via the process of fine-tuning hyper parameters and assessing the performance of the model. The Random Forest algorithm has exceptional performance in handling intricate datasets, offering significant insights into the importance of features.
- The current work employs a rigorous assessment technique to measure the efficacy of categorization classifiers. This is accomplished by the use of measurements such as F1 score, precision, and accuracy. In order to enhance the reliability of outcomes, researchers often use cross-validation methodologies, such as k-fold cross-validation. Examining the confusion matrix provides further information into the efficacy of the categorization algorithms. The comprehensive ten-step examination offers a comprehensive assessment of the capabilities of the models.

V. RESULTS AND DISCUSSION

We have looked at the KNN, RF and LR classification techniques. Results highlight how accurately the algorithms can sort based on certain characteristics. Choosing random forest is supported by the study's results. RF outperforms K-NN (65%) and LR (87%) with a precision rate of 94%, shedding light on its capabilities. It also sheds light on

importance of feature selection as picking the right attributes contributes to overall better performance. The higher precision rate of Rf implies suggests that it was successful in identifying important traits. Table 3 shows comparison in error values in the three algorithms: Random Forest, k-NN, and logistic regression. The proposed method is clearly more reliable as clearly indicated in Fig. 2.

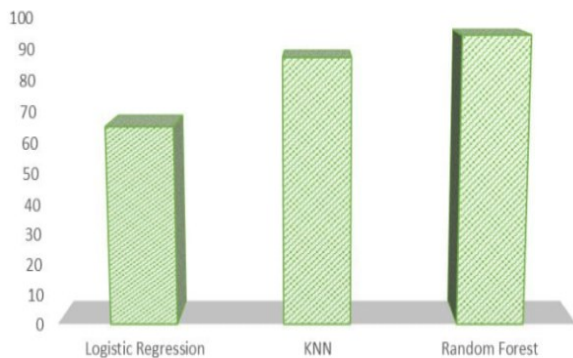


Fig.2. Accuracy Comparison

Further research is required to validate the data through large datasets. Investigating alternate feature selection techniques and measures of performance can provide valuable insights to CVD prediction. In conclusion, the usage of Random Forest shows great potential that can be explored.

TABLE 3: COMPARISON

Algorithms	MAE	R-Squared	MSE	RMSE
RF	0.237704918	0.530050216	0.117203287	0.342349644
KNN	0.237704918	0.177025862	0.205245902	0.453040728
LR	0.408214595	0.24504294	0.18828276	0.433915614

VI. CONCLUSION

In the years forward, classification algorithms can be upgraded by focusing on more advanced feature selection methods, incorporating deep learning, addressing imbalanced datasets and boosting efficiency and scalability. These improvements can account for more dependable and clear classifications, handling complex data patterns and lead to better forecasts. As a result, this study demonstrates how Machine Learning algorithms, in particular Random Forest, might improve the precision of heart disease prediction. The research emphasizes the potential for accurate early prediction of CVDs by utilizing a variety of evaluation approaches and feature selection strategies that are optimized. The findings of this study emphasize the significance of using advanced machine learning methodologies to enhance patient welfare and optimize medical diagnostic procedures.

REFERENCES

[1]. H. Sun and J. Pan, "Heart Disease Prediction Using Machine Learning Algorithms with Self-Measurable Physical Condition Indicators," *Journal of Data Analysis and Information Processing*, vol. 11, no. 01, pp. 1–10, 2023, doi: 10.4236/jdaip.2023.111001.

[2]. V. Chang, V. R. Bhavani, A. Q. Xu, and M. Hossain, "An artificial intelligence model for heart disease detection using machine learning algorithms," *Healthcare Analytics*, vol. 2, p. 100016, Nov. 2022, doi: 10.1016/j.health.2022.100016.

[3]. A. Abdellatif, H. Abdellatif, J. Kanesan, C.-O. Chow, J. H. Chuah, and H. M. Gheni, "An Effective Heart Disease Detection and Severity Level Classification Model Using Machine Learning and Hyperparameter Optimization Methods," *IEEE Access*, vol. 10, pp. 79974–79985, 2022, doi: 10.1109/access.2022.3191669.

[4]. G. N. Ahmad, H. Fatima, S. Ullah, A. Salah Saidi, and Imdadullah, "Efficient Medical Diagnosis of Human Heart Diseases Using Machine Learning Techniques With and Without GridSearchCV," *IEEE Access*, vol. 10, pp. 80151–80173, 2022, doi: 10.1109/access.2022.3165792.

[5]. G. N. Ahmad, S. Ullah, A. Algethami, H. Fatima, and S. Md. H. Akhter, "Comparative Study of Optimum Medical Diagnosis of Human Heart Disease Using Machine Learning Technique With and Without Sequential Feature Selection," *IEEE Access*, vol. 10, pp. 23808–23828, 2022, doi: 10.1109/access.2022.3153047.

[6]. K. C. Santosh et al., "Machine Learning Techniques for Human Age and Gender Identification Based on Teeth X-Ray Images," *Journal of Healthcare Engineering*, vol. 2022, pp. 1–14, Jan. 2022, doi: 10.1155/2022/8302674.

[7]. Jomy Mary Mathew, "HEART DISEASE PREDICTION AND COMPARATIVE ANALYSIS USING SUPERVISED MACHINE LEARNING ALGORITHMS," *International Research Journal of Modernization in Engineering Technology and Science*, Sep. 2022, doi: 10.56726/irjmet.30238.

[8]. P. Ghosh et al., "Efficient Prediction of Cardiovascular Disease Using Machine Learning Algorithms With Relief and LASSO Feature Selection Techniques," *IEEE Access*, vol. 9, pp. 19304–19326, 2021, doi: 10.1109/access.2021.3053759.

[9]. J. P. Li, A. U. Haq, S. U. Din, J. Khan, A. Khan, and A. Saboor, "Heart Disease Identification Method Using Machine Learning Classification in E-Healthcare," *IEEE Access*, vol. 8, pp. 107562–107582, 2020, doi: 10.1109/access.2020.3001149.

[10]. S. Mohan, C. Thirumalai, and G. Srivastava, "Effective Heart Disease Prediction Using Hybrid Machine Learning Techniques," *IEEE Access*, vol. 7, pp. 81542–81554, 2019, doi: 10.1109/access.2019.2923707.

[11]. S. P. Siddique Ibrahim and M. Sivabalakrishnan, "An Evolutionary Memetic Weighted Associative Classification Algorithm for Heart Disease Prediction," *Studies in Computational Intelligence*, pp. 183–199, Dec. 2019, doi: 10.1007/978-981-15-1362-6_9.

[12]. K. E. Akbas, M. Kivrak, A. K. Arslan, and C. Colak, "Assessment of Association Rules based on Certainty Factor: an Application on Heart Data Set," *2019 International Artificial Intelligence and Data Processing Symposium (IDAP)*, Sep. 2019, doi: 10.1109/idap.2019.8875977.

[13]. S. Bashir, Z. S. Khan, F. Hassan Khan, A. Anjum, and K. Bashir, "Improving Heart Disease Prediction Using Feature Selection Approaches," *2019 16th International Bhurban Conference on Applied Sciences and Technology (IBCAST)*, Jan. 2019, doi: 10.1109/ibcast.2019.8667106.

[14]. S. Kharya, S. Soni, and T. Swarnkar, "Weighted Bayesian Association Rule Mining Algorithm to Construct Bayesian Belief Network," *2019 International Conference on Applied Machine Learning (ICAML)*, May 2019, doi: 10.1109/icaml48257.2019.00013.

[15]. Satpathy, Raghunath. "Implementation of Some Bio-Inspired Algorithms in Prediction of Heart Diseases: A Review." *International Journal of Research in Electronics and Computer Engineering*, vol. 5, pp. 460–464, July–September 2017.